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Empirical nonergodic shaking scenarios based on spatial correlation models: An application to central Italy

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Abstract

This paper provides a new methodological framework to generate empirical ground shaking scenarios, designed for engineering applications and civil protection planning. The methodology is useful both to reconstruct the ground motion pattern of past events and to generate future shaking scenarios, in regions where strong-motion datasets from multiple events and multiple stations are available. The proposed methodology combines (1) an *ad-hoc* nonergodic ground motion model (GMM) with (2) a spatial correlation model for the source region-, site-, and path-systematic residual terms, and (3) a model of the remaining aleatory error to take into account for directivity effects. The associated variability is a function of the type of scenario generated (bedrock or site, past or future event) and it is minimal for source areas where several events have occurred and for sites where recordings are available. In order to develop the region-specific fully nonergodic GMM and to compute robust estimation of the residual terms, the approach is calibrated on a highly dense dataset compiled for the area of central Italy. Example tests demonstrate the validity of the approach, which allows to simulate acceleration response spectra at unsampled sites, as well as to capture peculiar physical features of ground motion patterns in the region. The proposed approach could be usefully adopted for data-driven simulations of ground shaking maps, as alternative or complementary tool to physic-based and stochastic-based approaches.

KEYWORDS

directivity effects, ground motion models, nonergodicity, shaking scenarios, spatial correlation

1 | INTRODUCTION

The generation of seismic shaking maps as a tool to support decision making at a given site is a key-topic for civil protection planning and engineering purposes. Currently, shaking fields are based on ground motion models (GMMs), which estimate the intensity measures as a function of several variables dependent on the reference earthquake scenario (magnitude, distance, soil category, etc). GMMs usually assume ergodic standard deviation[1], which implies higher level of uncertainty associated with the model predictions.

However, in case of site-specific applications (ie, loss assessment or risk analyses), the simplified assumption of *ergodicity* may be not reliable to describe regional properties of the ground shaking (ie, magnitude scaling, distance attenuation, and site effects). More accurate predictions are obtained through nonergodic approaches, in which the repeatable terms

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of variability are identified through statistical decomposition of the total residuals.[2] These terms are then used to correct the median GMM predictions, as well as to move part of the aleatory variability into epistemic uncertainty (eg, Anderson and Uchiyama, 2011 Refs. 3–8).

A common finding in the literature is that the residual terms of a GMM (the within-event part) are characterized by non-negligible spatial correlation,[9–11] which can be modeled through geospatial analysis and exploited to predict shaking intensities at sites where recording data are not available. As demonstrated by several authors, the spatial modeling is crucial to generate correlated ground motion random fields, so its neglecting may cause a remarkable bias in loss estimation and seismic risk analyses.[12–14] A similar principle is also applied to produce the USGS ShakeMaps,[15–17] an online tool where shaking maps are made available in near-real time for a discrete set of spectral ordinates. However, a limited number of studies in the literature considers fully nonergodic models and spatial correlation of the repeatable terms to generate ground motion fields. These few studies concern applications on well-sampled areas in California[18–21] and Europe.[8,22–24]

In this paper, we propose a methodology for generating data-driven shaking maps of acceleration spectral ordinates, based on the predictions of a nonergodic and region-specific GMM, in which the systematic contributions of variability are decomposed into event-, source-, site-, and path-effects. We develop spatial correlation models and interpolated maps for site- and path-terms, being the fundamental ingredients for generating adjusted scenario-based shaking fields. We also calibrate a model to take into account directivity-related effects, by modeling the azimuthal dependence of the remaining aleatory residuals.

On these bases, the proposed spatial correlation model allows reconstructing the regional ground motion pattern both of past and future scenarios, including the associated variability. The impact of the methodology is tested on the shaking fields and spectral intensity amplitudes simulated for the 2016 M_w 6.5 Norcia earthquake, as well as on an event independent of the dataset used to calibrate the model. The results show peculiar spatial patterns of the site and path effects in the central Italy, which are related to rheological and geological aspects of the study area.

2 | DATA

The dataset is composed of accelerometric and velocimetric high-quality earthquake signals, related to stations and events located in central Italy since 2008, thus, including data of the 2009 L'Aquila (M_w 6.1) and the 2016-2017 central Italy sequences (maximum magnitude M_w 6.5). It consists of more than 30,000 records of about 450 earthquakes in the magnitude range 3.4-6.5 and more than 460 stations within 120 km from the epicenters. This huge number of waveforms is due to the great effort made during the last seismic sequences in 2016 (Amatrice M_w 6.0—Visso M_w 5.9—Norcia M_w 6.5) by the Italian Department of Civil Protection (DPC), the Istituto Nazionale di Geofisica e Vulcanologia (INGV), and several academic and research institutions that installed more than 100 temporary stations with the aim of improving the event locations (Moretti et al., 2017)[25] and monitoring the site effects.[26,27] The same dataset was also adopted for other studies in the area, such as investigations on the temporal variability of ground motion,[28,29] microzonation studies,[26] as well as for calibrating a methodology aiming to identify the reference rock sites.[30] For more stable and robust analyses, we set the threshold to at least 10 recordings for each station, according to Lanzano et al.[7] The unselected stations were mainly located at the borders of the study area (additional details are available in Ref. 30).

Figure 1A depicts the magnitude-distance distribution of records in the dataset: the high density of events and stations in such relatively small area allows sampling a significant number of source to site ray-paths (Figure 1B), thus making this dataset particularly suitable to derive a fully nonergodic GMM.

3 | THE NONERGODIC GROUND MOTION MODEL

A region-specific GMM is here calibrated for central Italy to estimate the geometrical mean of the horizontal components of the IM-Intensity Measures (Peak Ground Acceleration, PGA and 5 acceleration response spectral ordinates, SA, within the period range 0.1-2 s). The model calibration is performed via a linear ordinary least-squares mixed-effect model31 (Bates, 2015). The functional form has been modified after Lanzano et al[30] so that the median prediction of IM is given by:

$$\widehat{\mu}_{es} = log_{10}(IM)_{es} = a + F_M(M) + F_R(M,R) + C_{es} + \delta W_0 \quad (1)$$

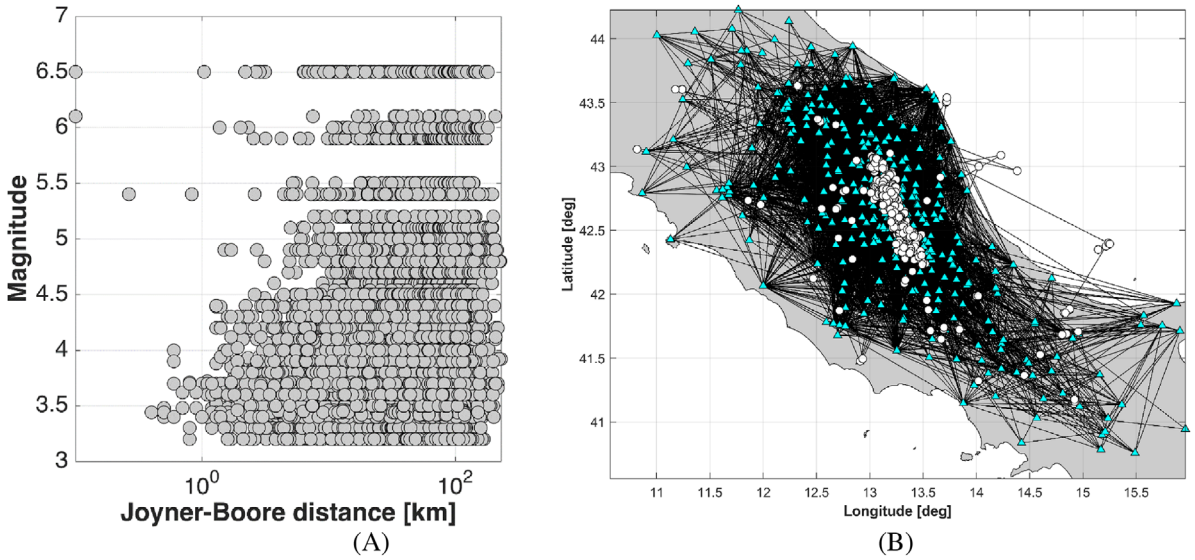


FIGURE 1 Magnitude-Joyner-Boore distance scatter plot of records in the dataset (A); map of the sampled ray-paths (black lines) from the events (circles) to the stations (triangles)

where a is the offset, while F_M and F_R are the magnitude and distance terms, respectively, which are expressed as:

$$F_M(M) = \begin{cases} b_1 (M_w - M_h) \; for \; M_w \leq M_h \\ b_2 (M_w - M_h) \; otherwise \end{cases}, \tag{2}$$

$$F_R(M,R) = \left[c_1 \left(M_w - M_{ref}\right) + c_2\right] \; log_{10} \frac{\sqrt{R_{JB}^2 + h^2}}{R_{ref}} + c_3 \left(\sqrt{R_{JB}^2 + h^2} - R_{ref}\right), \tag{3}$$

b_1, b_2, c_1, c_2, and c_3 are coefficients obtained from the regression; M_h (hinge magnitude), R_{ref} (reference distance), and h (pseudo-depth) are set to 5.0, 1 and 6 km, respectively; while the M_{ref} (reference magnitude) is obtained from a first-stage nonlinear regression. The distance metric adopted for this model is the Joyner-Boore distance, R_{JB}. The focal depth was not considered as an explanatory variable of the model, as the dataset refers to an homogeneous tectonic regime characterized by events with roughly constant depths (< 20 km), normally distributed $\sim N(9.3, 3.1)$ km.

In Equation (1), C_{es} is the corrective factor, including the systematic source, propagation, and site effects, depending on the earthquake scenario defined by the event e and site s. According to Stafford,[32] the corrective factor is introduced to remove the ergodic assumption and it is estimated using the random-effects on the systematic terms of events e (δB_e), stations s ($\delta S2S_s$), *source* ($\delta L2L_{source}$), and path p ($\delta P2P_p$), as follows:

$$C_{es} = \delta B_e + \delta S2S_s + \delta L2L_{source} + \delta P2P_p, \tag{4}$$

In Equation (4), δB_e represents the between-event (event-term), zero-mean, normally distributed residuals $\sim$N(0, τ_e), which correspond to the average misfit of recordings of one particular earthquake e with respect to the prediction of the fixed-effects GMM of Equation (1). The between-event corrections are available in Electronic Supplement #1 (ESUPP1).

The site-specific terms ($\delta S2S_s$) are also zero-mean normally distributed $\sim N(0, \phi_{S2S,s})$ random variables that define the systematic bias of ground motions recorded at each station and can be considered as a proxy of the empirical amplification of each station.[26,33] The $\delta S2S_s$ corrections are available in Electronic Supplement #2 (ESUPP2). In this case, we compute the site-specific terms with respect to the average GMM prediction provided by the fixed-effect model of Equation (1), which is calibrated on the reference stations (ie, rock sites unaffected by site-effects). These latter were previously identified on the study-area according to six proxies based on geophysical and seismological data, as well as on geomorphological and installation features.[30,34] The site-to-site terms at the reference sites are denoted by $\delta S2S_{ref}$ $\sim N(0, \phi_{S2S,ref})$, which are identified by a flag in Electronic Supplement #2 (ESUPP2).

TABLE 1 Coefficients and standard deviations of the proposed model

Model parameters	Intensity measures (IMs)					
	PGA	SA-T = 0.1 s	SA-T = 0.2 s	SA-T = 0.5 s	SA-T = 1 s	SA-T = 2 s
a	2.9118	3.0477	3.2293	3.2162	2.6114	1.9189
b_1	0.5450	0.5477	0.7133	0.8354	0.8464	0.7071
b_2	0.1925	0.1849	0.3237	0.4798	0.5245	0.4862
c_1	0.1809	0.1602	0.1009	0.1023	0.1488	0.2487
c_2	–1.4588	–1.3484	–1.3642	–1.5063	–1.3005	–1.2267
c_3	–5.766E-03	–8.015E-03	–5.309E-03	–1.645E-04	0	0
M_{ref}	3.8128	3.4098	3.5209	4.8719	5.4256	5.2690
τ_e	0.1527	0.1700	0.1449	0.1269	0.1393	0.1368
ϕ_{S2S}	0.2656	0.3033	0.2624	0.2439	0.2243	0.2010
$\phi_{S2S,ref}$	0.2011	0.2452	0.2049	0.1419	0.1153	0.1078
τ_{L2L}	0.0592	0.0705	0.0615	0.0157	0.0043	0
ϕ_{P2P}	0.0970	0.1063	0.0908	0.0755	0.0952	0.0984
σ_0	0.2103	0.2208	0.2241	0.1991	0.1842	0.1902
σ_{0d}	0.1585	0.1646	0.1671	0.1542	0.1531	0.1631

The source terms ($\delta L2L_{source}$) define normally distributed $\sim N(0, \tau_{L2L})$ systematic bias of the source regions (collected in Electronic Supplement #3—ESUPP3) associated with the GMM fixed-effects prediction, whereas the path-term ($\delta P2P_p$) defines the systematic deviation along one source-to-site path $\sim N(0, \phi_{P2P})$, which could be related to anisotropy in crustal velocity, density, or in the attenuation function.[6] The method of sampling, either for $\delta L2L_{source}$ and $\delta P2P_p$, is detailed in the next section.

Finally, δW_0 in Equation (1) denotes the residuals computed when the random-effects are removed from the total (logarithmic) difference between the observations and predictions. They are normally distributed with zero-mean and standard deviation σ_0, which represents the aleatory variability associated with the fully nonergodic model; its value is on average about 0.2 log10 units. The calibration results (coefficients and standard deviations) of the GMM are provided in Table 1 for the six IMs. The term σ_{0d} denotes the standard deviation of the model corrected for the azimuthal effects, as described later in the paper.

It is worth to be noted that the coefficient c_3, representing the amount of anelastic attenuation with distance, is typically regionally dependent. This parameter is a few orders of magnitude lower than the other model coefficients and shows strong sensitivity to small variations, that is, small changes of c_3 can lead to significant changes in the model predictions especially at long-distances. As a result, the attenuation term cannot be neglected, specifically at short-periods (ie, attenuation faster than the geometric one).

3.1 | Sampling of the corrective terms

The source terms $\delta L2L_{source}$ are identified according to a *clustering* approach, where the events are aggregated within polygonal source areas identified on the basis of space-time criteria (Figure 2). The Reasenberg's algorithm[35] is used to group the events: the earthquakes are counted within a time window of 10-days bins and considering a Poisson distribution. We obtain 20 clusters, which are further grouped based on the following criteria: (1) belonging to the same specific seismogenic setting (according to the source zones identified by DISS 3.2.1 catalogue, 2018), and (2) similarity in the stress drop[36,37] associated with the events of each cluster. In the end, six source areas are found (Table 2 and Figure 2A), each one identified by the corresponding main-event.

The most populated clusters are: *cluster* #1, including 2009 Mw 6.1 L'Aquila and 2017 Mw 5.5 Campotosto sequence, *cluster* #4, including 2016 Mw 6.0 Amatrice and Mw 6.5 Norcia sequences, and *cluster* #5, including 2018 Mw 4.6 Muccia sequence. The remaining not-clustered events are grouped as belonging to a *background* regional seismicity. As expected, being the considered sources originated from the same compression-extension tectonic setting,[38] the contribution of $\delta L2L_{source}$ is small, especially at longer periods.

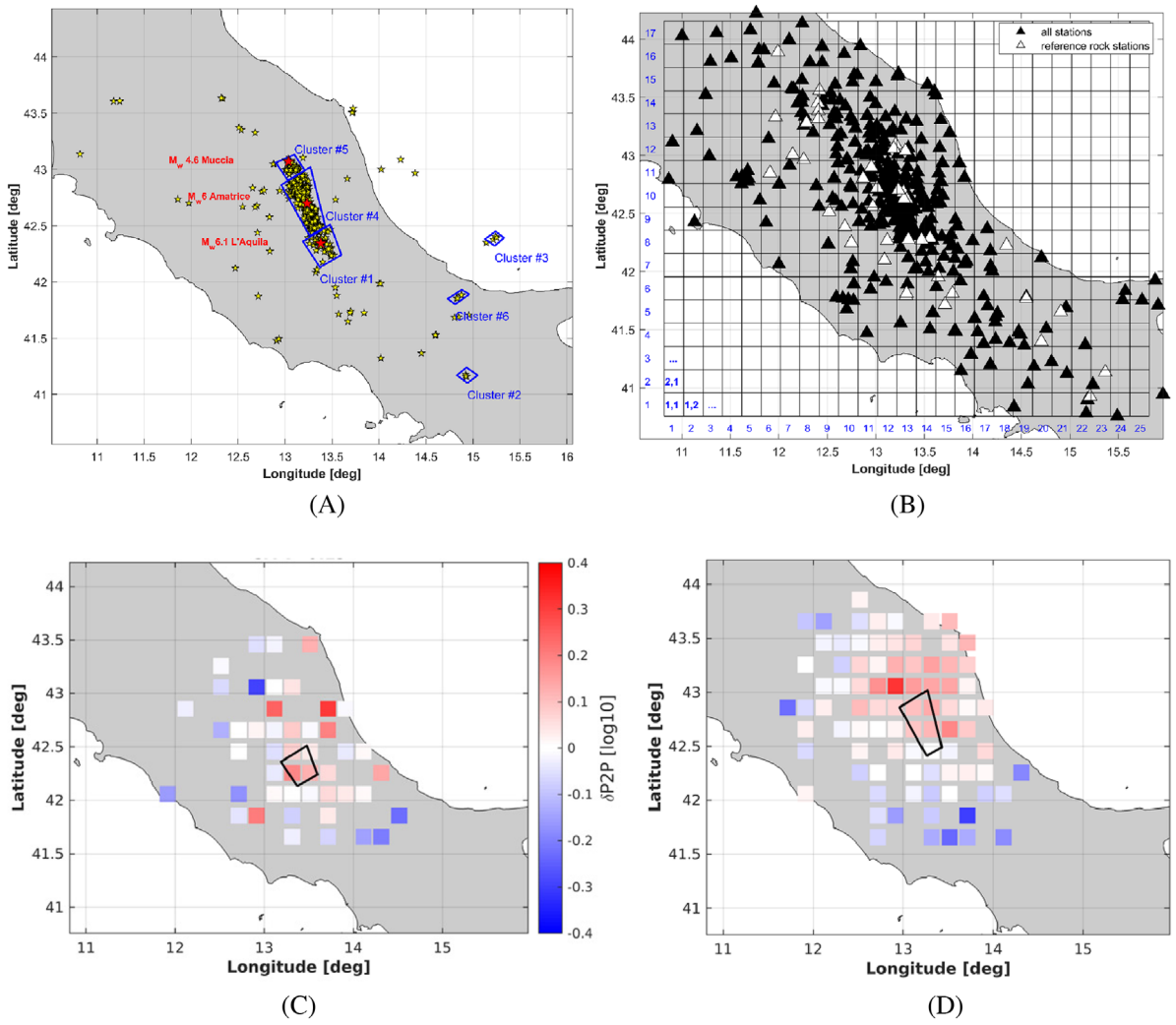


FIGURE 2 Source identification based on *clusters*: #1 (main event: L'Aquila 06/04/2009—01:32UTC), #4 (main event: Amatrice 24/08/2016—01:36UTC), and #5 (main event: Muccia 10/04/2018—03:11UTC) are considered in the present study (A); map of stations (black triangles) and reference rock sites (white triangles) overlapped to the grid with numbering of cells (blue text) (B); sampled paths from *cluster* 1 (C) and *cluster* 4 (D) at PGA

TABLE 2 Main features of the identified clusters

Cluster group	Reference DISS ID	Main-event	Average stress-drop [bar]
#CLUSTER1	ITCS013–ITCS040	M6.1, 6th April 2009	1.35
#CLUSTER 2	ITCS057	M4.2, 27th September 2012	–
#CLUSTER 3	ITCS052	M4.4, 6th December 2015	–
#CLUSTER 4	ITCS028–ITCS013–ITCS040	M6.0, 24th August 2016	1.92
#CLUSTER 5	ITCS028–ITCS056	M4.6, 10th April 2018	0.97
#CLUSTER 6	ITCS003–ITCS0059	M5.1, 16th August 2018	–

[a] Excluding the stress drop value of the main event from the average computing.

To account for the source-to-site paths $\delta P2P_p$, we divide the whole region into squared cells (0.2°-spaced), following the approach by Dawood and Rodriguez-Marek.[39] Then, we sample the paths starting from the *clusters* above defined (considered as source regions) to the sites (ie, the recording stations) identified on each cell of the regular grid. The mesh is depicted in Figure 2B with the corresponding cells numbering. It is composed of 425 cells in the latitude and longitude ranges [40.76, 44.36] and [10.82, 16.02], respectively (145 cells over 425 contained at least one station, for a total number of 288 sampled paths). As an example, Figures 2C and 2D show the sampled paths cell-by-cell from the source

(*clusters* 1 and 4, respectively), at PGA. The $\delta P2P_p$ corrections for *clusters* 1, 4, and 5 are available in Electronic Supplement #4 (ESUPP4).

4 | GEOSPATIAL ANALYSIS

As the main goal of the proposed methodology is to generate shaking maps, we need to compute corrected ground motion predictions not only at the recording stations but also at unobserved sites. To this aim, we estimate the path and site random terms, to be considered in Equation (4), from spatial correlation models (herein denoted as the *P-Model* and *S-Model*, respectively, for path and site corrections).

To evaluate spatial variability of the site and path residual terms, the traditional method of the semivariogram γ[40] is used, which describes the (semi)variance of the difference between a couple of values of spatially distributed random variables observed at the site s in the domain D (with s ∈ D) at a given pair of locations.

As commonly assumed in the practice,[41] among the others, we make the hypothesis of (space) stationarity and isotropy of the residual terms (ie, the correlation depends only on the separation distance between two pairs of sites), which follow a Gaussian normal distribution. An exponential model is selected for fitting the sample semivariograms by adopting the weighted least square technique:

$$\gamma\ (d) = a\left[1 - exp\left(-\frac{3d}{b}\right)\right], \tag{5}$$

in which a (*sill*) represents the semi-variance of the variable, b (*range*) is the separation distance d beyond which data are assumed to be spatially uncorrelated. For the exponential model, the *range* is conventionally defined as the separation distance d at which $\gamma(d)$ is equal to 95% of the *sill*. Based on the semivariogram models, a Kriging interpolation technique[42] is then applied as a predictor to estimate the value of the residual terms on the unsampled locations of a computation grid (herein assumed with 1.6 × 1.6 km resolution).

4.1 | The S-model

The geospatial analysis is applied on the residual site terms $\delta S2S_s$ to generate correlated fields of the local site effects (*S-model*). The *S-model* is characterized by a *lag distance* of the spatial correlation (ie, roughly equal to the half side of the square including all the considered stations—about 200 km—divided by a specified bin size) equal to about 3 km. The obtained values of the *range* and *sill* are plotted in Figures 3A and 3B, respectively. As it can be noted, the *sill* is constant (0.05 log10-units), while the *range* trend indicates relatively small correlation length (20 km on average) that shows a variable trend with a minimum at about 0.5 s.

Despite the correlation length is usually expected to proportionally increase with increasing period, a nonmonotonic trend between range and period was already observed in the literature (Esposito and Iervolino, 2012 among others).[43] Our findings, for instance, may be the result of the peculiar local site conditions that vary significantly across space, as expected for central Italy.[44] However, these results are consistent with the trend observed by Huang and Galasso[45] for Italian data and also confirm the findings by Schiappapietra and Douglas[14] on central Italy. These observations suggest that the rate of decay could be not only period dependent, but also regionally dependent, and thus a single model should not be used to describe the correlation features of different geographical areas.

The overall pattern of *S-model* is depicted in Figure 4A for SA(T = 1 s). In the figure, positive $\delta S2S_s$ indicate amplifications (red-color gradations), whereas negative values represent deamplifications (blue-color gradations) of ground-motion, with respect to the median prediction referred to rock sites. Variability associated with these terms was found to be relevant (~0.1 log10 units), when removed from the aleatory uncertainty.

A regional pattern is evident both at short and long periods, as the sites in the North-Eastern field of the study area (ie, the Marche region) show a clear amplification of the predicted ground motion, while in the Northern Apennines (Tosco-Emiliano) and in the central-southern area (Sannio-Irpinia), deamplifications are noticeable (Figure 4C). The prevalence of red color on the map is due to the adoption of a GMM calibrated on reference sites, so that the most of the points are affected by positive amplification with respect to the reference rock.

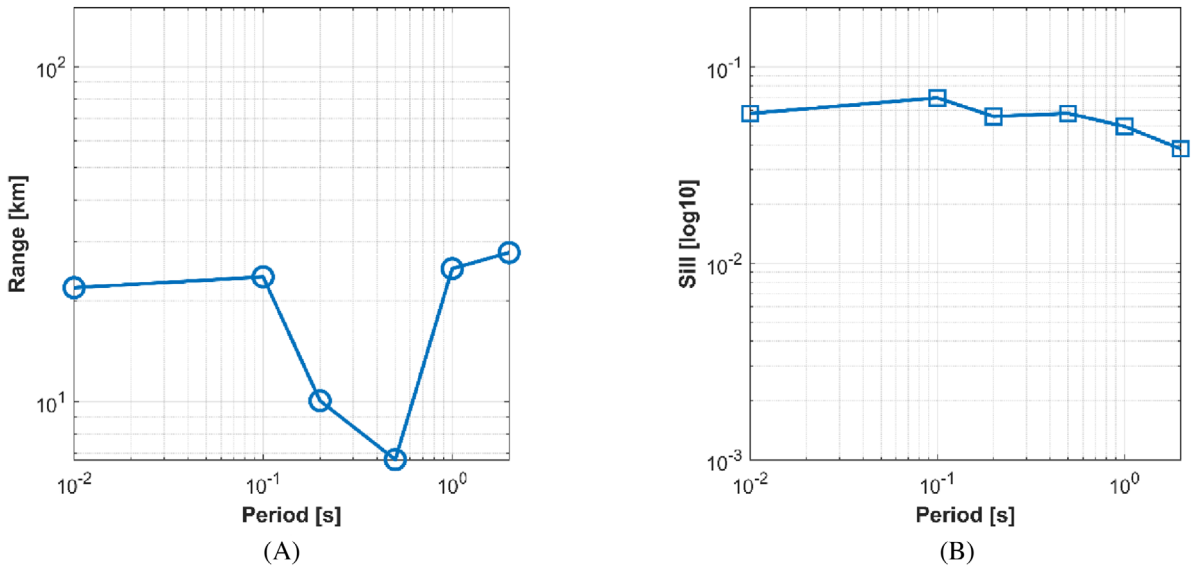


FIGURE 3 *Range* of correlation (A) and *sill* (B) as a function of the response spectral period T

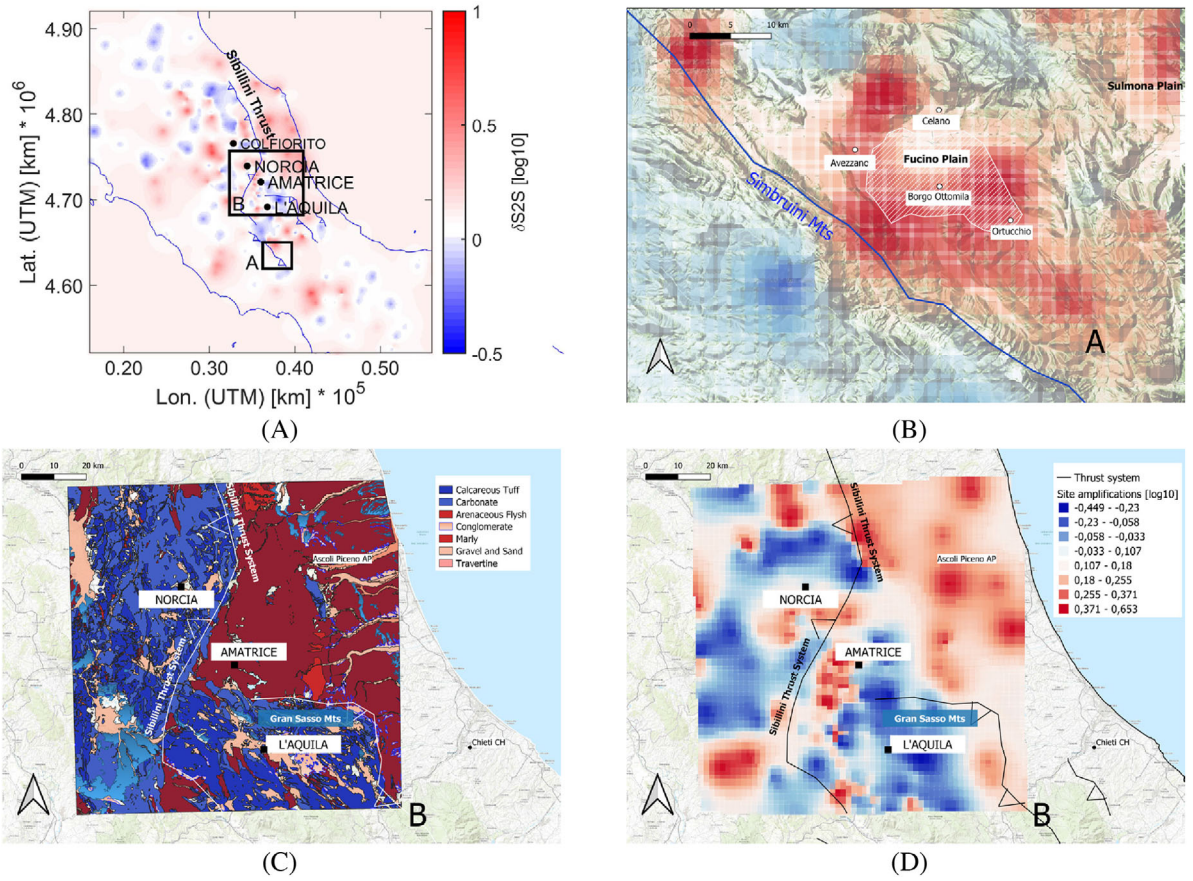


FIGURE 4 Example of spatially correlated fields of $\delta S2S_s$ site terms at SA(T = 1 s) – log10 units in central Italy: box A displays zoomed view on the Fucino basin (B); box B displays a zoomed view on the Amatrice area, with a simplified sketch of the main geo-lithological complexes (C), and the corresponding map of the site effects at SA(T = 1 s) (D)

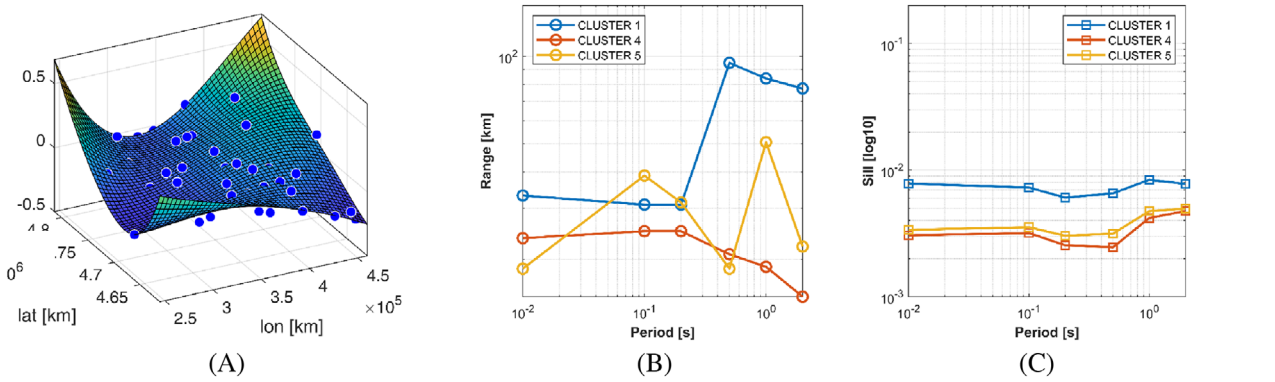


FIGURE 5 Example of interpolating surface trend of the path terms $\delta P2P_p$ for #*cluster* 1 at PGA (A), *range* of correlation (B), and *sill* (C) on logarithmic scale

Despite the complex geological setting of the region, which is characterized by the presence of the central Appenines together to basins and soft sediments, the major features are captured by the $\delta S2S_s$ model that is densely constrained by the wide covering of recording stations in the area. The observed spatial trends in $\delta S2S_s$ could be potentially related to local heterogeneities of the site conditions, mainly dependent on the shallow soil layers. Details at larger geological scale allow to appreciate these aspects (zoomed views A and B of Figure 4A): as an example, in Figure 4B, the site effects at SA(T = 1 s) around the Fucino plain are mapped. The chart is overlapped on DEM at 250 m, showing the main geomorphological characteristics of the intramountains plain. It can be noted that more enhanced effects (red colored) are concentrated in the basins (ie, Fucino and Sulmona), in correspondence of softer deposits that in this area typically show amplification at longer periods (≥1 s).

Another example is in Figure 4C, where a detail of the geo-lithological complexes around the well-sampled municipality of Amatrice is shown. The area is characterized by the presence of foredeep basin (between the Sibillini thrust and the Gran Sasso Mts front), covered by softer flysch sediments (ie, Ref. 46), where we expect larger site amplifications compared to more compact lithological types lying on the footwall of the Sibillini thrust and at Southern of Gran Sasso. These geological-based considerations may potentially play a prominent role in the physical interpretation of the $\delta S2S_s$ spatial pattern, particularly at the period T = 1 s. Indeed, a roughly consistent distribution of the site amplifications on the hanging wall side of the Sibillini thrust toward the Adriatic foreland can be observed in Figure 4D.

4.2 | The P-model

The path residual terms of each *cluster* are spatially modeled ($\delta P2P_p$). Resulting maps of these terms are denoted as *P-model* in the rest of the paper. In this case, the assumed *lag distance* is about 5 km. A relevant finding of the geostatistical analysis of the path terms relates to the presence of a trend (ie, a spatially nonconstant mean), that is clearly highlighted by an unstable *sill* of the variogram. This evidence of nonstationarity is dealt with a detrending processing of the residuals before applying the variogram model and the Kriging algorithm. In detail, a fourth-order polynomial function of the geographic coordinates is considered to remove the trend (Figure 5A). In this way, the interpolation model of $\delta P2P_p$ is regarded as the sum of two contributions—Equation (6): (1) a deterministic and spatially varying component (ie, the drift,$m_s$with s ∈ D − Equation 7) and (2) a stationary stochastic component of detrended residuals δ_s

$$\delta P2P_p(s) = m_s + \delta_s, \tag{6}$$

$$m_s = \sum_{l=0}^{L} a_l f_l(s), \tag{7}$$

where, for l = 0,…,L, a_l are scalar coefficients and $f_l(s)$ are known functions, which play the same role as regressors in the linear model framework. The resulting covariance structure is characterized by a stable *sill* (Figure 5C) and a *range*

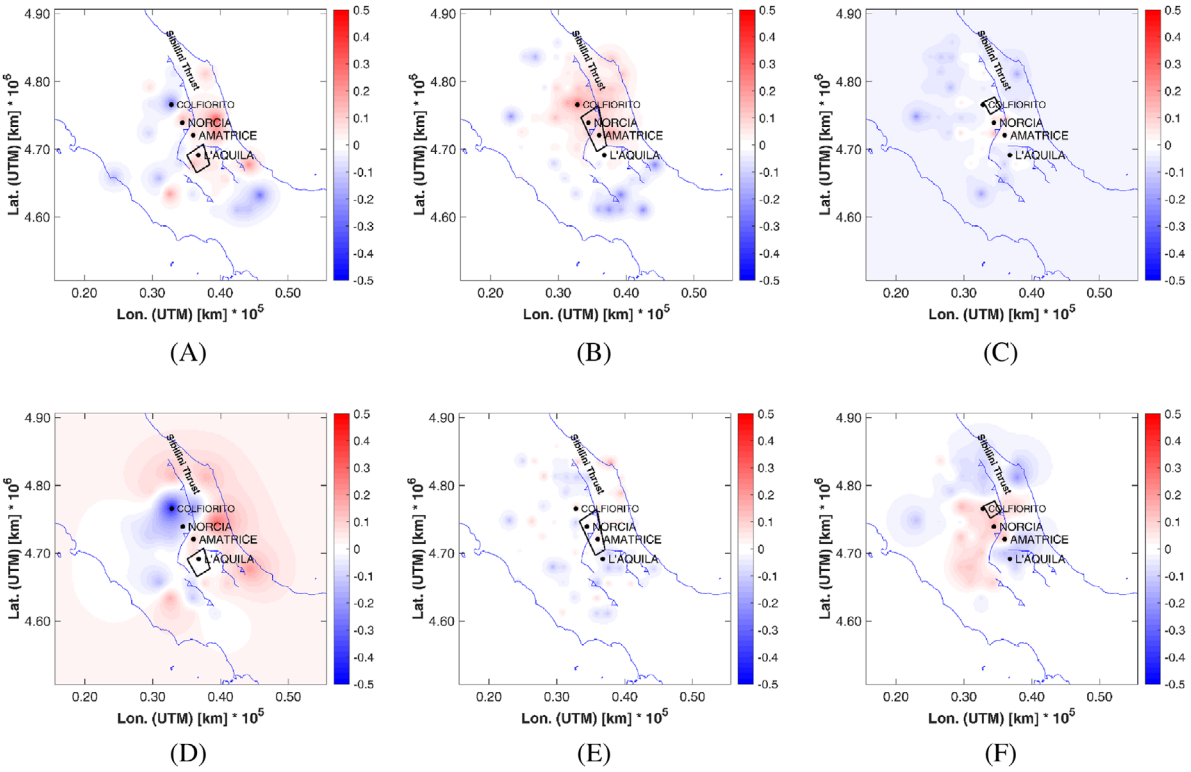


FIGURE 6 Example of spatially correlated fields (*P-Models*) for L'Aquila region—#*cluster* 1 (A), Amatrice-Norcia region—#*cluster* 4 (B), and Colfiorito-Muccia region—#*cluster* 5 (C) at PGA and SA(T = 1 s), respectively (D-F). Black polygons indicate the cluster areas. Lines indicate the main thrust systems of the region

(Figure 5B) that increases up to 80–100 km with increasing periods for #*cluster* 1 (L'Aquila source), whereas a roughly constant/decreasing trend is observed at about 20–40 km for the other *clusters*.

Spatial pattern of the $\delta P2P_p$ terms depends on the location of the source-region as observed in Figure 6 (log10 values) for three investigated *clusters* and for two example periods (PGA—top panel and SA(T = 1 s)—bottom panel).

The maps of Figure 6 show remarkable path effects in the region: for L'Aquila source area (#*cluster* 1), the spatial distribution of the residuals at PGA (Figure 6A) and SA(T = 1 s) (Figure 6D) is positive for the seismic ray-paths traveling from the Apennine chain to the Adriatic coast (red-colored contours), whereas deamplifications are observed toward the Tyrrhenian coast (blue-colored contours) and Southern regions. An opposite pattern can be recognized for #*cluster* 5 (Figure 6C, F), while #*cluster* 4 exhibits an intermediate behavior with amplification focused at Northern of the source area (Figure 6B, E).

These observations reveal that path effects may be affected by the relative position of the source with respect to the main discontinuity structure of the region. The latter is represented by the Sibillini (extensional) thrust system, which separates the whole central Italy region into two halves: the western side dominated by Mesozoic carbonatic structures and the eastern side by post-Messinian foredeep deposits.[46] For L'Aquila source (#*cluster* 1), for instance, the ray-paths leave from the hanging wall of the main thrust fault and then propagate toward the Eastern coast, where contribute to amplify the ground motion. Conversely, for #*cluster* 5, located on the footwall of the main fault system, amplifications occur toward S-SW direction, on the same side of the source with respect to the thrust system; #*cluster* 4 instead straddling the two side of the Sibillini fault, leading to an intermediate behavior with respect to the previous ones, with strong Northern amplification that tends to vanish with period increasing (Figure 6E).

The results depicted in Figures 4 and 6 appear justified by physical considerations, even if misleading interpretations may rise from potential trade-off between correlated site ($\delta S2S_s$) and path terms ($\delta P2P_p$). Evidence of trade-off in GMM fixed- and random-effects is indeed well known and expected, as also highlighted, for instance, by Baltay et al[47] for source and path residuals of a nonergodic model. In this regard, we perform some tests to check if the randomization of the repeatable terms is effective in removing the trade-off among the residual components. The statistical test is based on the

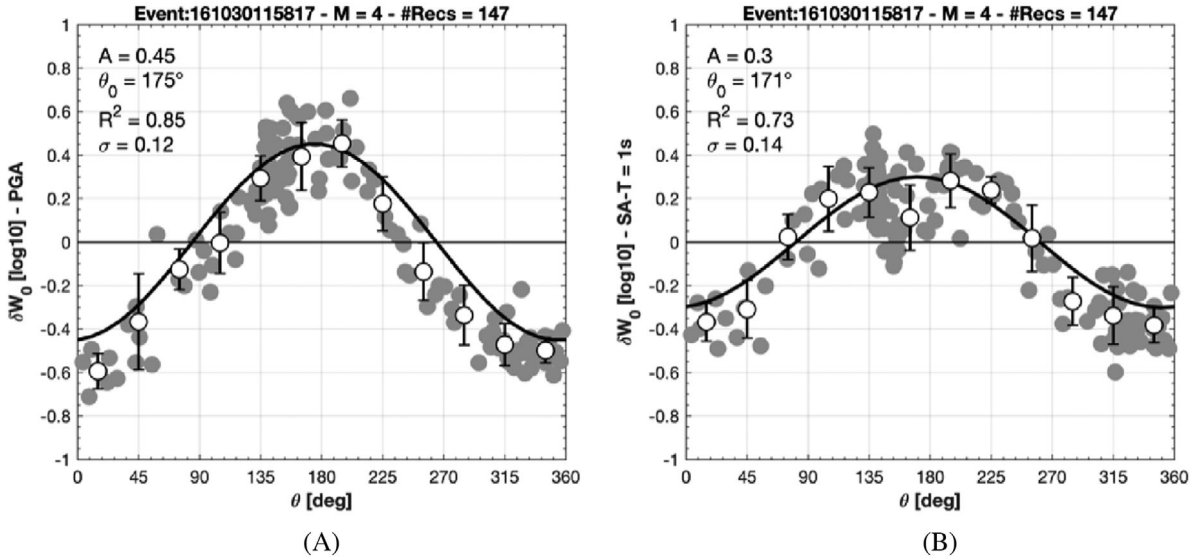


FIGURE 7 Azimuthal variation of the aleatory residuals fitted with Equation (8) for the event M4, recorded on 30.10.2016 11:58:17 at PGA (A) and SA(T = 1 s) (B)

Pearson coefficient,[48] which grounds on the assumption that trade-offs between pairs of the considered random terms are quantified by the off-diagonal elements of the variance-covariance matrix.

For the sake of brevity, we report the results in the Supplementary Material. Here, we limit to discuss that, as far as the correlation coefficient concerns, we find a relatively small trade-off between site and path terms in all the considered *clusters*, except around the period 0.1 s where correlation increases. The trade-off is limited, so the main findings about the reported maps on site and path effects may still be valid.

4.3 | The D-model

Rupture-directivity effect is one of the main factors in determining the spatial pattern of seismic shaking, generating the largest values when the sites are located in the forward-direction with respect to the rupture propagation. While these trends are well recognized for strong events, recently, they have been also observed for small earthquakes.[49–52]

In the following, according to the approach by Luzi et al,[53] we build a fitting model to take into account for the azimuthal variation of the residuals corrected for the repeatable effects. To this aim, we fit the aleatory residuals δW_0 of each event in the dataset as a function of the source-to-site azimuth θ. The fitting model is a simple cosinusoidal function, selected on the basis of the trend of residual distribution with θ:

$$\delta D2D = A\cos(\theta - \theta_0), \tag{8}$$

where A is the amplitude of the fitting cosine function and θ_0 is the azimuth at the maximum amplitude; in detail, source-to-site azimuth θ is calculated counterclockwise from zero at geographical north. The values of A and θ_0, calibrated event-by-event over δW_0, are reported in Electronic Supplement #5 (ESUPP5) and #6 (ESUPP6), respectively.

As an example, Figure 7 illustrates the event Mw 4.0, occurred on 30 October 2016 (aftershock of M_w 6.5 Norcia mainshock) on which a strong directional effect is observed, more pronounced at short-periods (T = 0.1 s; Figure 7A) than at long-periods (T = 1 s; Figure 7B). Empirical evidence of the period-dependency of directivity effects on small-to-moderate earthquakes was indeed found by some authors,[52] by fitting the residuals obtained removing path and site effects.

A statistical analysis is performed on the $\delta D2D$ parameters, estimated for more than 400 events in the dataset. Figure 8 shows the histograms of amplitudes A and θ_0, highlighting the contribution of the events with significant directivity effects, thus characterized by a statistical coefficient of determination R^2>0.5. As it can be noted in Figure 8A, the amplitudes A at PGA of the events with R^2>0.5 (yellow-colored histogram) show a log-normal distribution with the median equal to 0.225 log10 units, which corresponds to an average of 1.7 amplification factor. As far as θ_0 is concerned

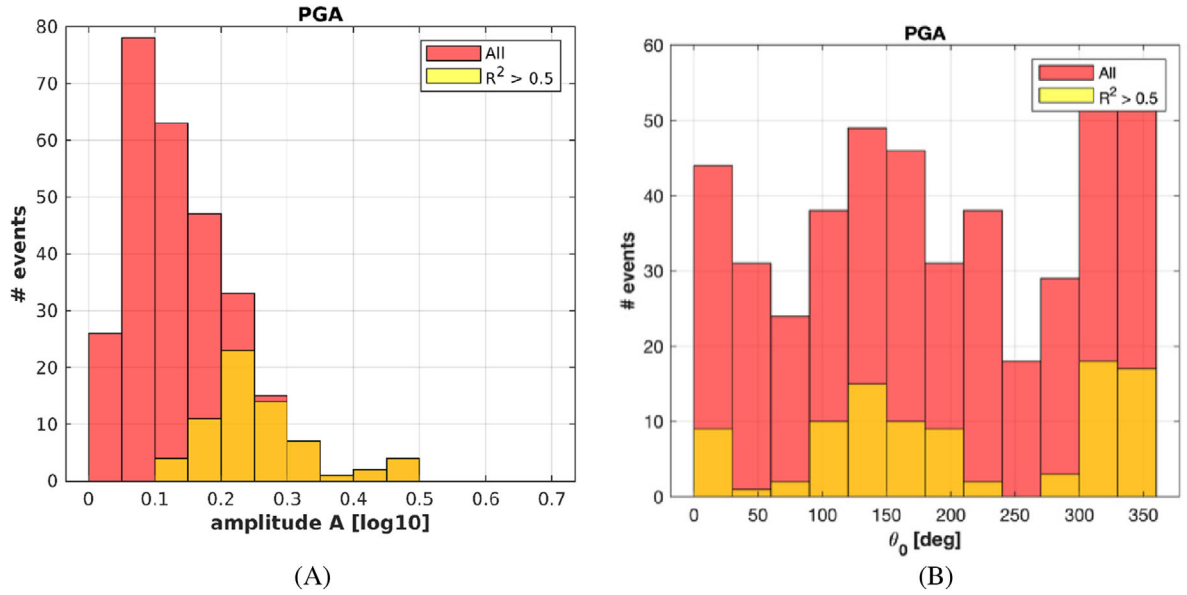


FIGURE 8 Statistical distribution of $\delta D2D$ parameters at PGA: amplitudes A (A) and θ_0 (B), related to all the events in the dataset (red-colored histogram) and to the subset of the statistically significant (R^2>0.5) events (yellow-colored histograms)

TABLE 3 Mean amplitude μ_A of $\delta D2D_m$ for different magnitude ranges [log10]

M range	# events	PGA	SA-T = 0.1 s	SA-T = 0.2 s	SA-T = 0.5 s	SA-T = 1 s	SA-T = 2 s
All	456	0.1203	0.1267	0.1319	0.1093	0.0848	0.0859
M<3.6	197	0.1218	0.1309	0.1326	0.0952	0.0719	0.0762
3.6< = M<4.1	182	0.1161	0.1235	0.1295	0.1151	0.0868	0.0865
4.1< = M<5.0	62	0.1310	0.1276	0.1417	0.1401	0.1137	0.1060
M> = 5.0	15	0.1105	0.1085	0.1154	0.1271	0.1664	0.1618

(Figure 8B), the distribution is bimodal with two peaks around 150° and 330°, corresponding to the average direction (toward the South and North, respectively) of the fault strikes of the central Apennines events (Mw>5.5).

As a result, the statistical analysis reveals that about 30% of the events shows remarkable azimuthal dependence, with preferential direction aligned along the Apennine chain.

Table 3 reports the median amplitude μ_A of all the events grouped by different magnitude ranges. We find that directivity depends not only on period but also on magnitude (ie, magnitude-dependent directivity correction denoted by $\delta D2D_m$): for larger events (M>4.5), greater amplifications are expected to occur as long as the period increases, whereas small events show larger median amplifications at shorter periods.

We also find that the residual aleatory uncertainty σ_{0d} (ie, corrected for the site, path, source, and directivity effects) is almost constant over all periods and roughly equal to 0.155 (log10 units)—see the corresponding values in Table 1.

5 | WORKFLOW FOR SIMULATING SHAKING FIELDS

All the terms estimated in the previous analysis are the ingredients to generate shaking scenarios in terms of median values and associated variability. Technically, the procedure combines (logarithmically) the various terms of Equation (1) at each node of the regular grid, depending on the input scenario. The relationships to obtain the median prediction and the associated standard deviations are detailed in the Appendix, where we distinguish for 16 different simulating-scenario options (identified by a SIM number), which are defined according to the following criteria:

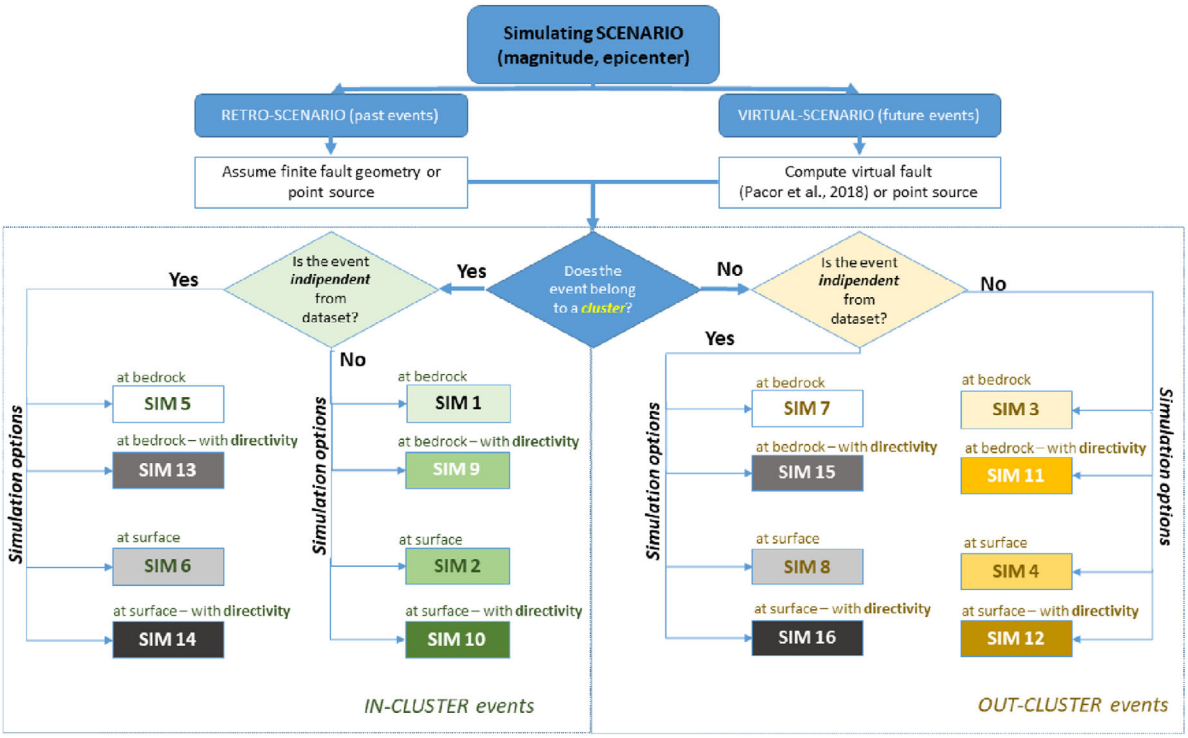


FIGURE 9 Workflow of the empirical simulation of seismic scenarios. The median prediction and the associated standard deviation for each simulating option (SIM) are detailed in the Appendix

- Simulating-scenarios of events belonging to one of the six identified source-*clusters* (ie, falling in the *clusters* with known path correction $\delta P2P_p$) or falling in the *background* seismicity (ie, without path correction). In the latter case, the path variability ϕ^2_{P2P} is accounted for in the model standard deviation;
- Simulating-scenarios of events included in the calibration dataset (ie, with known between-event terms δB_e) or not included, thus assuming zero-mean between event terms. In the latter case, the between-event variability τ^2_e is accounted for in the model standard deviation;
- Simulating-scenarios of ground motion at surface (ie, including the site correction $\delta S2S_s$) or at bedrock (ie, without site correction). In the latter case, the variability of the reference sites $\phi^2_{S2S,ref}$ is accounted for in the model standard deviation;
- Simulating-scenarios of ground motion with directivity effects (ie, including the azimuthal correction $\delta D2D$ for events inside the calibration dataset or the predicted $\delta D2D_m$ for scenarios outside the calibration dataset) or without directivity effect (ie, without the azimuthal correction). In the latter case, the variability of the azimuthal correction $\phi^2_{D-model}$ is accounted for in the model standard deviation.

A schematic workflow of the overall methodology is illustrated in Figure 9. In the following, we describe two examples of shaking scenarios: the first one corresponding to a past event belonging to the dataset (2016 M_w6.5 Norcia earthquake) and the second one to a smaller event occurred in central Italy in 2019, which is not included in the calibration dataset.

The adoption of different schemes for empirical simulation, as those proposed in the Appendix, allows to cover different types of scenarios. As an example, events located outside the assumed source clusters and not included in the dataset maybe simulated at bedrock though SIM-option 7 (see Appendix). In these cases, the median shaking predictions come down to the simple median of the GMM, without any adjustment of event, path, site, and directivity terms. Hence, the corresponding systematic effects, which are not included in the median prediction through the correlated random fields, are taken into account in the model variability σ. The same scenarios at the surface (SIM-option 8), instead, are obtained by summing the median GMM at bedrock and the *S-model*, which, however, is well sampled in the whole domain, so that we do not expect anomalous patterns in ground motion.

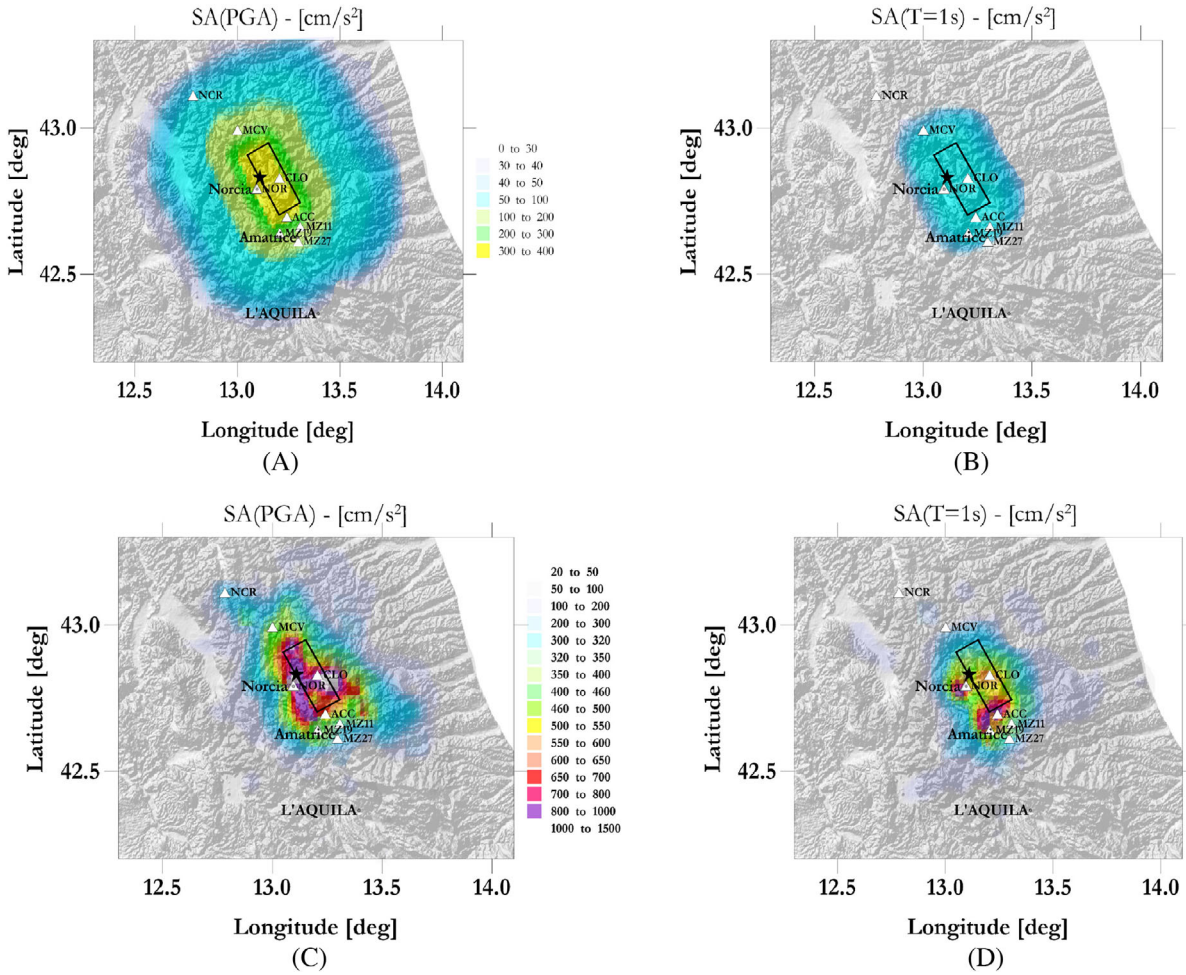


FIGURE 10 Shaking pattern of Mw 6.5 Norcia earthquake (median field) for PGA (A) and SA(T = 1 s) (B) at bedrock and for PGA (C) and SA(T = 1 s) (D) at surface. Triangles indicate the recording stations considered for the aims of comparison. Black star denotes the epicenter of the event; fault is plotted as a rectangle

6 | THE M_w6.5 NORCIA CASE STUDY

The following case study focuses on the ground motion fields produced by the earthquake M_w6.5 occurred in Norcia on 30 October 2016 at 06:40:18 UTC (magnitude, location, and fault geometry are taken from ITACA database; http://itaca.mi.ingv.it/).[54] The reference *cluster* of this event is #4. The amplitude A of the corresponding $\delta D2D$ function (as reported in supplement ESUPP5) shows a maximum value at T = 1 s with θ_0 equal to 174°, indicating that in this case, the azimuthal effect is maximal along the strike direction (NW-SE) of the fault toward SSE, which is an effect potentially related to rupture directivity.

Maps of the ground motion at bedrock (median field) are thus obtained by applying the proposed methodology (*SIM* 1 and *SIM* 10 of Appendix) at each node of the computing grid, as an example for PGA (Figure 10A) and SA(T = 1 s) (Figure 10B), respectively. The spatial distribution of intensity shaking, including the azimuthal correction, is shown at PGA (Figure 10C) and SA(T = 1 s) (Figure 10D).

The obtained maps produce reasonable estimates of ground motion spatial distribution, which is able to capture the shaking pattern focused on the SW margin of the fault projection, as observed by other authors (eg, Ref. 53). The corresponding maps of variability are shown in Figure 11 for PGA. Uncertainty, indeed, spatially depends both on the Kriging standard deviation of the *P-model* ($\beta_{P\text{-}model}$)—for simulation at bedrock (*SIM* 1 of the Appendix)—and on the *S-model* ($\beta_{S\text{-}model}$)—for simulation at surface (*SIM* 10 of the Appendix). As it can be noted, uncertainty map at bedrock is almost constant over the space domain, with smaller values in correspondence of the centroid of each cell considered for the paths sampling (Figure 11A), according to the pattern of $\beta_{P\text{-}model}$.

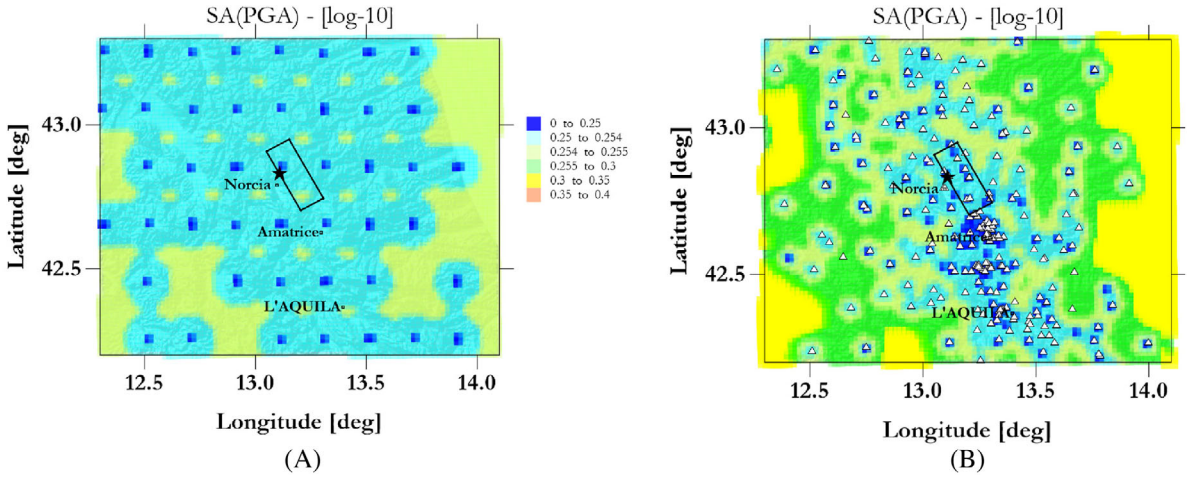


FIGURE 11 Uncertainty maps of the Mw 6.5 Norcia shaking scenario for PGA at (A) bedrock and (B) surface. Triangles indicate the recording stations considered for model calibration. Black star denotes the epicenter of the event; the surface projection of the fault is plotted as a rectangle

For simulations at surface (Figure 11B), the variability map shows minimum values in correspondence of the recording sites, where the Kriging variance of the *S-model* is zero, thus the plotted standard deviation reduces to the aleatory variability of the GMM, when corrected for all the systematic effects (σ_0).

Spectral amplitudes (SAs) plotted in Figure 12 for stations selected above or in the proximity of the fault are compared with the recorded data, along with the median prediction of the model ITA18.[55] The root mean squared error (RMSE) is provided for each station to test goodness of the predictions.

As shown in this example, the proposed model provides improved predictions than ITA18; however, it is worth to be noted that the dataset adopted herein and the one used to calibrate ITA18 are remarkably different, so that the ITA18 (regional) model cannot be considered the "ergodic version" of the (local) GMM proposed in this paper. Therefore, the obtained results cannot be extended to more general findings about the comparison between nonergodic versus ergodic models.

Figure 13 depicts the boxplot of the model error with periods, computed as arithmetic mean over 177 recording stations. The average performance (RMSE = 0.18524) is good as demonstrated by the comparison with the ITA18 (mean RMSE = 0.2539) and the proposed model without the directivity effects (mean RMSE = 0.1996).

Some biased results ~0.1 (log10) at the longest period (T = 2 s) are likely due to source effects only partially captured by the model (for instance, related to near-source effects). Nonetheless, overall, the simulated spectra appear in good agreement with the actual observations.

7 | APPLICATION ON AN INDEPENDENT EVENT

The empirical model is herein used to generate shaking scenarios of an independent event (ie, not included in the dataset used for calibration) occurred in central Italy (M_l 4.2, 01/09/2019) and recorded by 137 stations (data are taken from the Engineering Strong-Motion database ESM, http://www.esm.mi.ingv.it).[56] As this event is characterized by small magnitude, the fault geometry is not defined and we assume a point source.

We apply the nonergodic model at each node of the computing grid according to Equation (1) and compute the remaining source, site, and path residuals (δW_0). As it can be noted from Figure 14A, the testing event is characterized by anisotropic spatial distribution of δW_0. Positive bias (ie, observations larger than predictions with average residuals ~0.2 log10 units) is observed at shorter periods for stations approximately located toward NW direction (azimuth ~ 300°-330°); conversely, negative residuals are noticeable along the opposite direction (azimuth ~ 120°-150°), thus, suggesting that directional features affect the ground motion. We interpret these trends as rupture-directivity effects due to the following considerations: (1) they are independent on distance; (2) the direction of maximum amplification is mainly NW-SE, coherently with the alignment of the central Italy fault systems (strike-parallel direction); and (3) they persist at high frequency, beyond the corner frequency of the event.[52,57]

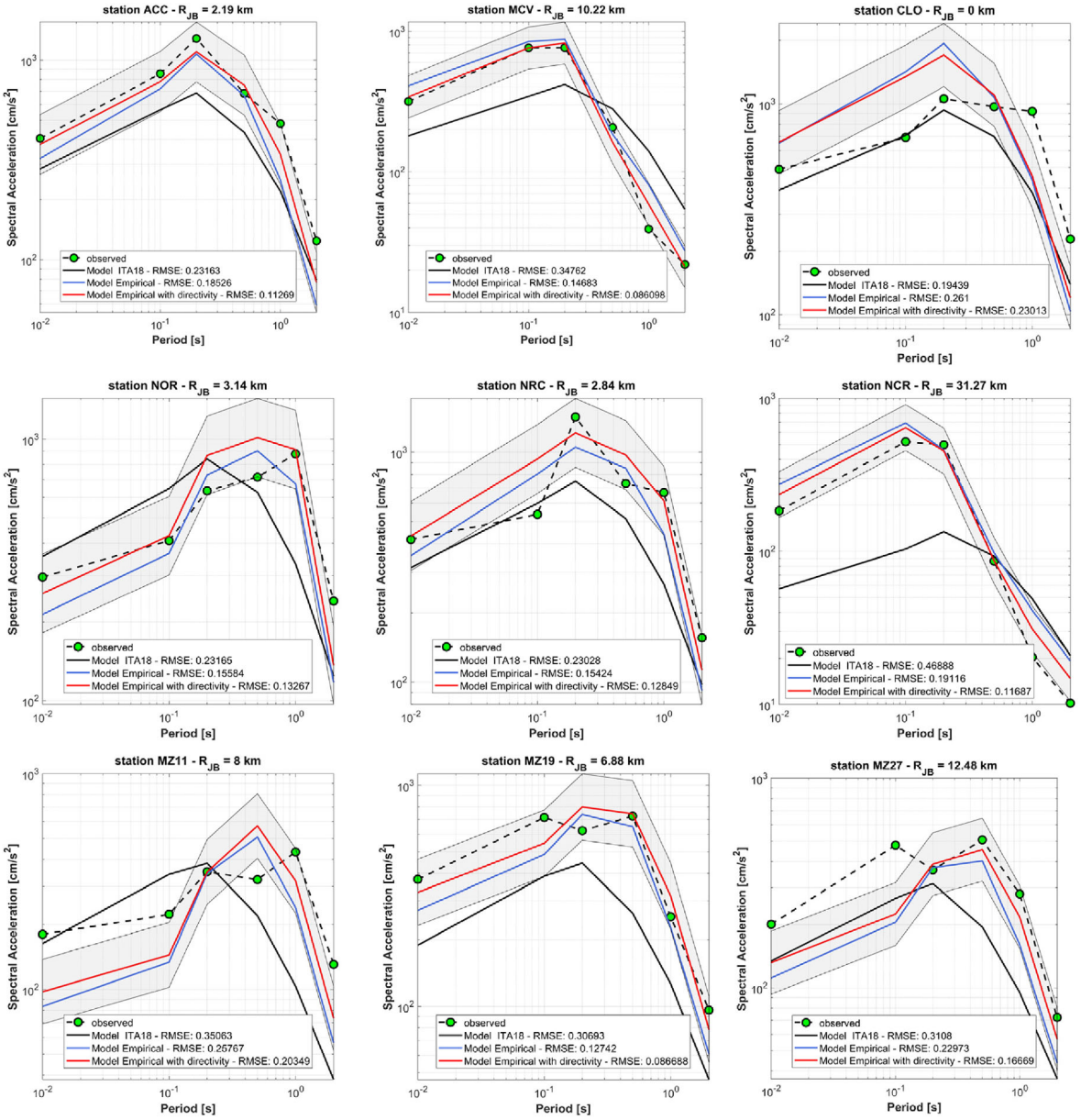


FIGURE 12 Comparison between observed spectral values (green circles) at six periods, ITA18 (black lines), spectral predictions by the proposed model uncorrected for directivity effects (blue lines), and corrected for directivity effects (red lines), along with the corresponding plus/minus standard deviation (gray band around the median), for selected stations triggered by the M_w6.5 Norcia 30.10.2016 event: ACC—MCV—CLO (top panel), NOR—NRC—NCR (middle panel), and MZ11—MZ19—MZ27 (bottom panel). The lines are obtained by interpolation of the considered (six) spectral ordinates

As we have no *D-model* data for this event, we extract the azimuthal corrections to be used in Equation (8) from the statistical distributions of the amplitudes A and azimuth θ_0 (as shown in Figure 8).

In detail, the median amplitude μ_A is assumed according to the values of $\delta D2D_m$ reported in Table 3 for the corresponding magnitude bin, whereas the azimuth θ_0 is assumed equal to 300° at all periods, as suggested from visual inspection of δW_0 spatial distribution. After *D-model* correction, the average shaking pattern at PGA (Figure 14B) is computed according to the relationships of *SIM 14* (Appendix).

Figure 15 displays SA at some selected stations near the epicenter, as predicted through the proposed empirical model, compared to the observed values at the same stations, along with ITA18 predictions.

The empirical model improves the average prediction over all stations of about 10% in terms of mean RMSE (0.228), against to the predictions obtained without directivity correction.

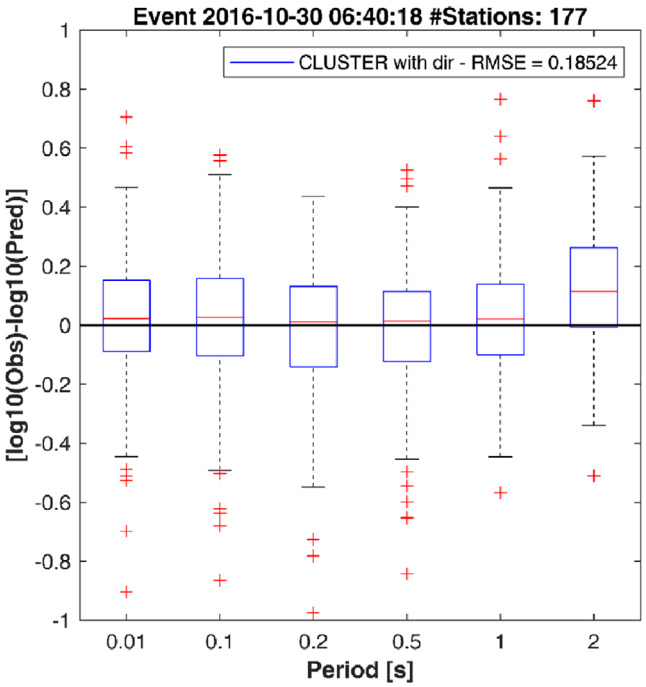


FIGURE 13 Boxplot for the event M_w6.5 Norcia 30.10.2016. T = 0.01 s is used for plotting PGA

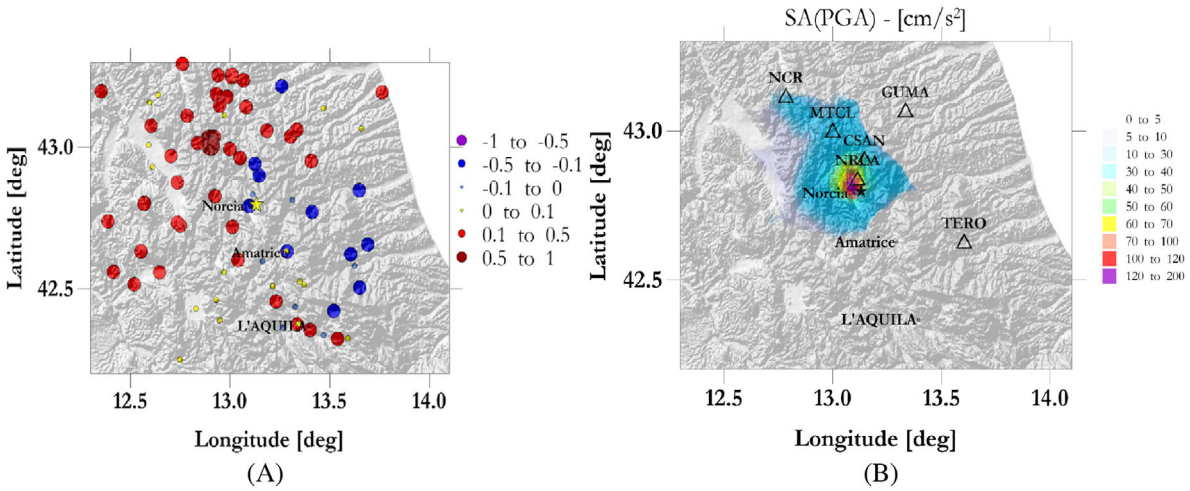


FIGURE 14 Spatial distribution of PGA of source-, site-, and path-corrected residuals $\delta \boldsymbol{W}_0$ around the epicenter with red-color graded circles indicating positive residuals and blue-color graded circles indicating negative residuals (A); shaking pattern of Ml4.2 Norcia earthquake (median field) for PGA (B). Triangles indicate the recording stations; the star marker denotes the epicenter and the fault is plotted as a closed rectangle

To analyze the efficiency of the methodology in predicting the observed SAs, the Pearson statistical correlation coefficient $\rho_{pred-obs}$ is computed for all stations distant less than 100 km from the epicenter. Results are reported in Table 4.

It can be observed that the proposed model produces appreciable improvement at all periods compared to ITA18, particularly at short periods, as the empirical model better captures possible systematic effects related to the source, path, site, and directivity. This application demonstrates the ability of the method to reproduce shaking scenarios in a predictive framework. Inconsistencies detected on specific sites, particularly at period 0.1 s, may be dependent on effects related either on source (eg, nonsystematic bias caused by the slip distribution, rupture nucleation, etc) or on site and path, not fully captured by the model.

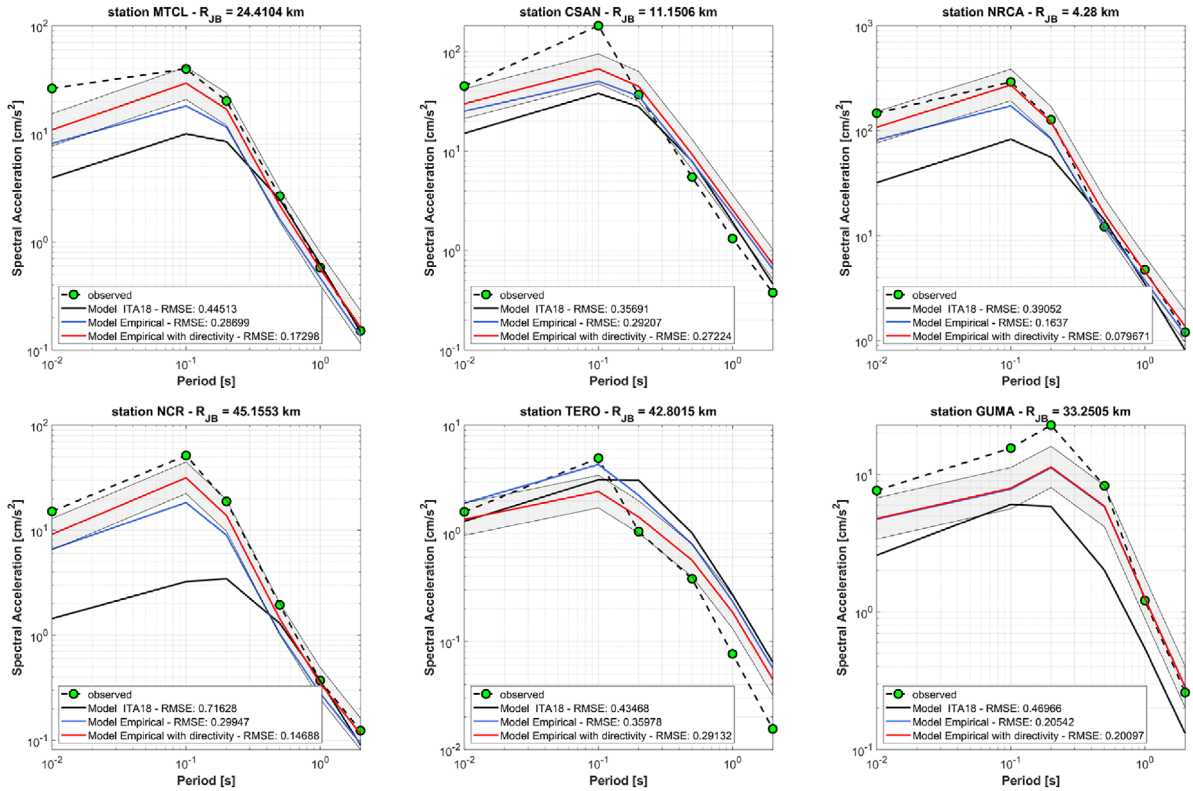


FIGURE 15 Comparison between observed spectral values (green circles), ITA18 (black lines), spectral predictions by the proposed models with (red lines), and without directivity effects (blue lines) along with the corresponding plus/minus standard deviation (gray band around the median) for some stations triggered by M_l 4.2 Norcia 01.09.2019 event: MTCL—CSAN—NRCA (top panel), NCR—TERO—GUMA (bottom panel)

TABLE 4 Pearson statistical correlation coefficient $\rho_{pred-obs}$ for M_l4.2 Norcia 01.09.2019 event

	PGA	SA(T = 0.1 s)	SA(T = 0.2 s)	SA(T = 0.5 s)	SA(T = 1.0 s)	SA(T = 2.0 s)
ITA18	0.7180	0.5941	0.8345	0.8007	0.7251	0.8374
Empirical model	0.8897	0.7575	0.8909	0.8387	0.7973	0.8579
Empirical model with D-model correction	0.9051	0.7585	0.8896	0.8464	0.8167	0.8729

8 | CONCLUSIONS

In this paper, we have proposed a novel methodology to empirically simulate shaking scenarios for central Italy. The strategy of simulation is based on the combination of a region-specific and fully nonergodic GMM with the spatial correlated fields of the systematic residual terms for source, site, and paths. A further corrective term of the median ground motion is also incorporated in the algorithm to properly account for directivity effects, which also implies a drastically reduction of the aleatory variability, to about 0.15 log10-units at all periods.

On a practical level, the proposed method would apply in case of earthquake occurrence in central Italy (with M≥3.4) to adjust the region-specific GMM predictions with the correlated random fields. The adjustment terms may be also declined as a function of the scenario configuration (according to the operative scheme given in the Appendix), and provided along with a map of the associated variability.

Basically, the application-oriented potential of the methodology is mainly intended for:

- generating earthquake-induced *shaking fields*, so as to obtain a first quasi-real time processing of the ground motion pattern. The maps may be also continuously updated (ie, adjusting the correlated random fields of the corrective terms), as new records become available;
- reproducing scenarios of potential *future events*, thus useful in region-specific risk analysis and loss models;

- post-processing shaking maps of *past earthquakes*, wherever poorly constrained by a limited number of sampled records;
- designing an intermediate-level and prompt technique for *scenarios simulation*, useful until the results of more complex numerical models are made available.

The provided examples have shown that the model predicts the observed spectral intensity fairly well, specifically near and above the surface fault projection. The method also better captures the local heterogeneities of shaking due to geological and topographic features of the area, compared to global GMMs or standard ergodic approaches, which rather use proxies that are more limited in predicting site/path effects.

Further developments are ongoing, including calibration of advanced *D-model* to incorporate dependency on magnitude explicitly, as well as the parametrization of the *S-model*, in order to constrain the correlation model to specific proxies of local site response. This should also allow to provide more effective separation of the site contribution, thus, potentially resolving the observed trade-off with the path terms at short periods.

Future works will be devoted to assessing the impact of the methodology in the framework of site-specific hazard analysis and to extend the approach to less-sampled regions of Italy, for instance, by lowering the threshold magnitude to increase the number of sampled events.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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Appendix

List of options for simulating scenarios

Symbols:	
μ_{es}	Prediction of fixed-coefficients ground motion model
δB_e	Between-event correction
$\delta S2S_s$	Site-to-site correction (from S-model)
$\delta L2L_{source}$	Location-to-location correction
$\delta P2P_p$	Path-to-path correction (from P-model)
$\delta D2D$	Directivity effects correction for events in the dataset (D-model)
$\delta D2D_m$	Directivity effects correction from magnitude-dependent model (D-model)
σ_0	Aleatory variability event-, site-, and path-effects corrected
σ_{0d}	Aleatory variability event-, site-, path-, and directional effects corrected
τ_e	Variability of the between-event term
τ_{L2L}	Variability of the location-to-location term
ϕ_{P2P}	Variability of the path-to-path term
ϕ_{S2S}	Variability of the site-to-site term
$\phi_{S2S,ref}$	Variability of the site-to-site term of reference sites
$\beta_{P-model}$	Kriging variability of the P-model
$\beta_{S-model}$	Kriging variability of the S-model
$\phi_{D-model}$	Variability of the D-model

SIM OPTIONS	Median ground motion prediction $\hat{\mu}_{es}$ [log10]	Standard deviation σ [log10]
1. Simulation at bedrock of an IN-CLUSTER event, included in the calibration dataset	$\mu_{es} + \delta B_e + \delta L2L_{source} + \delta P2P_p$	$\sqrt{\sigma_0^2 + \phi_{S2S,ref}^2 + \beta_{P-model}^2}$
2. Simulation at surface of an IN-CLUSTER event, included in the calibration dataset	$\mu_{es} + \delta B_e + \delta L2L_{source} + \delta P2P_p + \delta S2S_s$	$\sqrt{\sigma_0^2 + \beta_{P-model}^2 + \beta_{S-model}^2}$
3. Simulation at bedrock of an OUT-CLUSTER event, included in the calibration dataset	$\mu_{es} + \delta B_e$	$\sqrt{\sigma_0^2 + \phi_{S2S,ref}^2 + \tau_{L2L}^2 + \phi_{P2P}^2}$
4. Simulation at surface of an OUT-CLUSTER event, included in the calibration dataset	$\mu_{es} + \delta B_e + \delta S2S_s$	$\sqrt{\sigma_0^2 + \beta_{S-model}^2 + \tau_{L2L}^2 + \phi_{P2P}^2}$
5. Simulation at bedrock of an IN-CLUSTER event, not-included in the calibration dataset	$\mu_{es} + \delta L2L_{source} + \delta P2P_p$	$\sqrt{\sigma_0^2 + \phi_{S2S,ref}^2 + \tau_e^2 + \beta_{P-model}^2}$
6. Simulation at surface of an IN-CLUSTER event, not-included in the calibration dataset	$\mu_{es} + \delta L2L_{source} + \delta P2P_p + \delta S2S_s$	$\sqrt{\sigma_0^2 + \tau_e^2 + \beta_{P-model}^2 + \beta_{S-model}^2}$
7. Simulation at bedrock of an OUT-CLUSTER event, not-included in the calibration dataset	μ_{es}	$\sqrt{\sigma_0^2 + \phi_{S2S,ref}^2 + \tau_e^2 + \tau_{L2L}^2 + \phi_{P2P}^2}$
8. Simulation at surface of an OUT-CLUSTER event, not-included in the calibration dataset	$\mu_{es} + \delta S2S_s$	$\sqrt{\sigma_0^2 + \beta_{S-model}^2 + \tau_e^2 + \tau_{L2L}^2 + \phi_{P2P}^2}$
9. Simulation at bedrock of an IN-CLUSTER event, included in the calibration dataset, with directionality	$\mu_{es} + \delta B_e + \delta L2L_{source} + \delta P2P_p + \delta D2D$	$\sqrt{\sigma_{0d}^2 + \phi_{S2S,ref}^2 + \beta_{P-model}^2}$
10. Simulation at surface of an IN-CLUSTER event, included in the calibration dataset, with directionality	$\mu_{es} + \delta B_e + \delta L2L_{source} + \delta P2P_p + \delta S2S_s + \delta D2D$	$\sqrt{\sigma_{0d}^2 + \beta_{P-model}^2 + \beta_{S-model}^2}$
11. Simulation at bedrock of an OUT-CLUSTER event, included in the calibration dataset, with directionality	$\mu_{es} + \delta B_e + \delta D2D$	$\sqrt{\sigma_{0d}^2 + \phi_{S2S,ref}^2 + \tau_{L2L}^2 + \phi_{P2P}^2}$
12. Simulation at surface of an OUT-CLUSTER event, included in the calibration dataset, with directionality	$\mu_{es} + \delta B_e + \delta S2S_s + \delta D2D$	$\sqrt{\sigma_{0d}^2 + \tau_{L2L}^2 + \phi_{P2P}^2 + \beta_{S-model}^2}$
13. Simulation at bedrock of an IN-CLUSTER event, not included in the calibration dataset, with directionality	$\mu_{es} + \delta L2L_{source} + \delta P2P_p + \delta D2D_m$	$\sqrt{\sigma_{0d}^2 + \tau_e^2 + \phi_{S2S,ref}^2 + \beta_{P-model}^2 + \phi_{D-model}^2}$
14. Simulation at surface of an IN-CLUSTER event, not included in the calibration dataset, with directionality	$\mu_{es} + \delta L2L_{source} + \delta P2P_p + \delta S2S_s + \delta D2D_m$	$\sqrt{\sigma_{0d}^2 + \tau_e^2 + \beta_{P-model}^2 + \beta_{S-model}^2 + \phi_{D-model}^2}$
15. Simulation at bedrock of an OUT-CLUSTER event, not included in the calibration dataset, with directionality	$\mu_{es} + \delta D2D_m$	$\sqrt{\sigma_{0d}^2 + \tau_e^2 + \phi_{S2S,ref}^2 + \tau_{L2L}^2 + \phi_{P2P}^2 + \phi_{D-model}^2}$
16. Simulation at surface of an OUT-CLUSTER event, not included in the calibration dataset, with directionality	$\mu_{es} + \delta S2S_s + \delta D2D_m$	$\sqrt{\sigma_{0d}^2 + \tau_e^2 + \tau_{L2L}^2 + \phi_{P2P}^2 + \beta_{S-model}^2 + \phi_{D-model}^2}$